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Sub - Cloud computing

Handwritten project Reflection → Assignment III

1) what I implemented ?

→ In this project, I have developed & deployed a light weight command parsing language model using Knowledge distillations. I used phi-3.5-mini-struct as a teacher model to generate structured command-parsing responses & these teacher-student generated examples where then used to train an EleutherAI/pythia-70m student Model.

I trained the student model using 1500 teacher-generated examples & evaluated it on the provided validation dataset. The final model achieved 99% quality / JSON validity successfully meeting the required the 99% quality gate. After training the model I deployed it as a fast API service using dockers & Kubernetes. I used fast/API service using docker & kubernetes. I used 'SeaweedFS' as S3-Compatible object storage so that the model could be stored seperately from the application image & downloaded with the application started.

3) problem I faced.

One of the major problems I encountered during training was undesirable repetitive & Buggy behaviours in an early configurations. The model sometimes produced repeated or prematurely terminated outputs. I addressed this by using the complete sequence for the causal language model objective, retraining from a base model & using a more conservative learning rate.

I also fixed configure issued IPs, synchronize the trained model files to S3 - compatible storage, configure kubernetes secrets & make sure that the model could download & load the model correctly during startup.

4. future improvements:

I will continue to working on this projects. First priority would be reducing inference latency; Techniques such as model quantization & inference optimization could reduce CPU & Memory requirements & potentially improve responsiveness.

I would also work on improving the models semantic accuracy, particularly its full-form accuracy.

This helped me understand how a Machine learning model can be treated as a versioned deployment artifact rather than being directly embedded into the application.

what I learned?

This project helped me understand the complete lifecycle of a ml application, starting from dataset generation & model training & continuing through deployment, testing, monitoring & recovery

I learned how different kubernetes components work together. In particular, I gained practical experience with deployment, services, Ingress, readiness probes, liveness probes & Horizontal pod Auto Scaling (HPA).

I also learned that deployment a ml model is significantly different from simply running the model locally. In a real deployment factors such as model storage, startup time, memory & CPU requirements, health checks, request routing & failure recovery all become important.

The CI/CD implementation was another important learning experience. I learned how model evaluation can be incorporated into an automated pipeline rather than treating model quality as a separate model step.